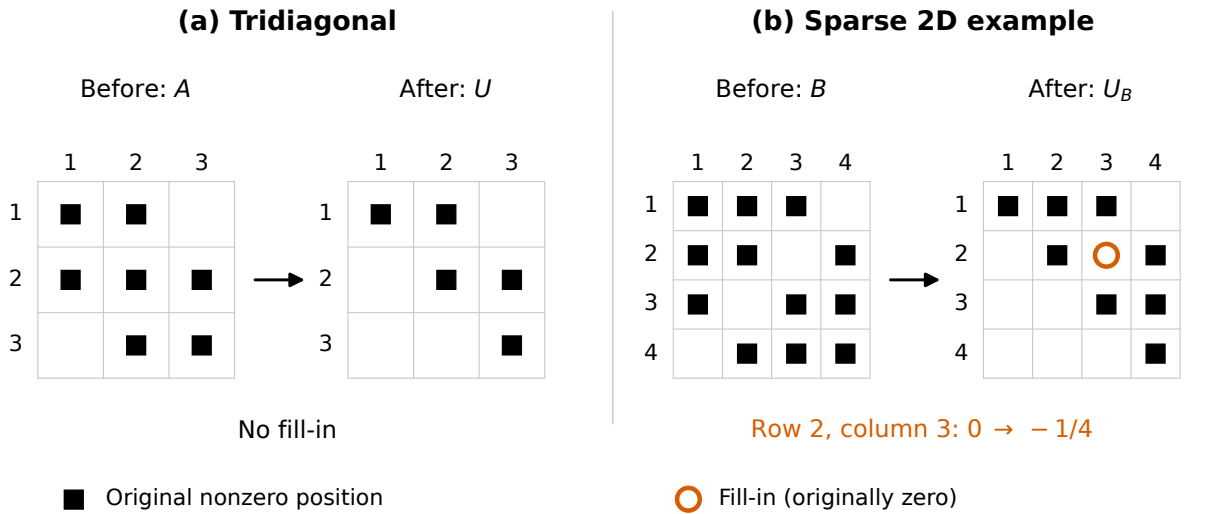
(a) Tridiagonal
Before: A
After: U
1 2 3
1
2
3
No fill-in
(b) Sparse 2D example
Before: B
After: U_B
1 2 3 4
1
2
3
4
Row 2, column 3: 0 → − 1/4
Original nonzero position
Fill-in (originally zero)
Blank cells = zero. Gauss elimination without row swaps.